

# Charmi Padh

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## SUMMARY

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Computer Engineering undergraduate and AI/ML developer with experience in real-time intelligent systems and scalable ML solutions. Industry experience through internships in deploying practical AI applications. Research contributions include paper presentations and multiple works under review in computer vision, deepfake detection, and predictive analytics. Finalist in multiple hackathons, demonstrating strong problem-solving and technical expertise.

## EDUCATION

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**LDRP Institute of Technology and Research, KSV** Gandhinagar, Gujarat  
Bachelor of Engineering in Computer Engineering 2023-2027  
CPI: 8.35 (Till Semester 5)

**12<sup>th</sup>, Shri P. V. Modi School** Jamnagar, Gujarat  
Cumulative Percentage: 65% Jul 2022 - Apr 2023

## WORK EXPERIENCE

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**Reliance Industries Limited** 14th May 2025 – 14th Jun 2025  
*Role: AI/ML Intern* 

- Developed a gasifier refractory lifespan prediction system by integrating and preprocessing IP21 and LIMS datasets to analyze degradation patterns and estimate the remaining useful life of refractory bricks.
- Developed advanced CNN-LSTM models with feature/time attention to capture refractory wear patterns and analyze degradation patterns and estimate the remaining useful life of refractory bricks.

**Institute for Plasma Research (IPR)** 9th Dec 2025 – 27th May 2026  
*Role: AI/ML Project Intern* 

- Developed and implemented offline reinforcement learning algorithms using pre-collected datasets for decision-making without real-time environment interaction.
- Designed and optimized model pipelines to improve policy learning efficiency and stability in data-constrained scenarios.

## PROJECTS

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**Intelligent Traffic Optimization System**   2025  
Real-Time Traffic Monitoring System - *Frontend Developer*

- Built a real-time traffic monitoring web app using FastAPI, React (Vite), and YOLOv5s trained on the IITM-HeTra dataset.
- Achieved 71% detection accuracy with 1.4 ms inference time through an optimized AI pipeline for efficient traffic analysis.

**Prompt-Injection-Prevention**  2025  
AI Safety System for LLMs - *Machine Learning Developer*

- Built a hybrid AI safety system combining rule-based filters and a Vision Transformer (ViT) classifier to detect unsafe prompts with 98.32% accuracy.
- Trained and optimized a ViT-based model on labeled prompt datasets to identify harmful patterns and enhance LLM security and reliability.

**InnovAITion Hackathon 2026**  Dec 2025 - Jan 2026  
Entropy: Real-Time Fraud Detection - *Machine Learning Developer*

- Designed a multi-step real-time fraud detection pipeline using XGBoost and LSTM for accurate transaction analysis.
- Achieved 4th runner-up position among 185 teams, demonstrating strong performance in large-scale competitive settings.

## RESEARCH PROJECTS & PUBLICATIONS

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### TrafficEye : Intelligent Traffic Optimization using Deep Learning Approach



Published in *IEEE International Conference on Artificial Intelligence and Machine Vision (AIMV 2025)*

- Built a YOLOv5s-based traffic optimization system (71% accuracy, 1.4 ms inference) and optimized the pipeline for adaptive Traffic signal control.

### Research Book Chapter

Accepted

*Springer Book Chapter*

- Co-authored a Springer book chapter on quantum simulation tools, evaluating platforms like Qiskit, Azure Quantum, and Cirq for architecture, performance, and limitations.

### PharmaADR-AI : A Pharmacovigilance-Driven Explainable Ensemble Machine Learning Framework for Opioid Adverse Drug Reaction Prediction

Under Review

*Elsevier : Computational Toxicology Journal*

- Proposed an explainable ensemble framework leveraging pharmacovigilance data for accurate prediction and interpretation of opioid-induced adverse drug reactions.

### SecurePark : Real Time Edge AI and IOT Based Vehicle Parking Ecosystem

Under Review

*International Conference on Converging Intelligence - 2026*

- Designed a real-time edge AI and IoT-based smart parking system for efficient vehicle detection, space utilization, and traffic optimization.

### SecureSamvaad: A Multimodal Fusion Approach For Real-Time Telecom Fraud Detection

Under Review

*International Conference on Converging Intelligence - 2026*

- Developed a multimodal fusion framework integrating audio, text, and metadata for robust real-time telecom fraud detection.

### From Theory to Practice: A Survey and Case-Based Analysis of Face-Swapping Deepfake Detection Models

Under Review

*MDPI : MultiMedia Tools and Applications Journal*

- Conducted a comprehensive survey and case-based evaluation of face-swapping deepfake detection models, highlighting practical challenges and performance gaps.

## LEADERSHIP & Volunteer EXPERIENCE

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### Short Term Training Program (STTP) on Mathematics-Driven Machine Learning

Dec 2024

#### • Student Volunteer

Short Term Training Program (STTP) on Mathematics-Driven Machine Learning: Concepts, Techniques, and Applications, organized by the IEEE LDRP-ITR Student Branch; contributed to the successful execution of expert-led sessions on core machine learning concepts.

### Digital & Design Committee Member

Aug 2024 – Jan 2025

IEEE LDRP-ITR Student Branch



- Created digital and print assets for IEEE events using Canva, Adobe Illustrator, and Figma, enhancing event visibility and student engagement through creative design thinking.

## SKILLS & CERTIFICATIONS

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### Skills:

- **Programming Languages:** C, C++, Python, HTML5, CSS
- **Frameworks & Libraries:** Flask, Django, Tailwind CSS, Pandas, NumPy, TensorFlow, Keras, OpenCV, FastAPI
- **Databases:** MySQL, PostgreSQL
- **Tools & Technologies:** Git, GitHub, Power BI

### Certifications:

- NPTEL Data Analysis & Algorithms (Elite)
- Conference Paper Presentation at IEEE AIMV 2025

## Achievements:

- **Finalist – Tic-Tech-Toe 2024**

Organized by Dhirubhai Ambani University; developed “EduMate” for enhanced classroom interaction.



- **Research Grant**

SecurePark — Edge AI, vision, and IoT-based campus security and parking ecosystem; awarded Rs. 69,017 by MMPSRPC, KSV.



- **Conference Grant**

TrafficEye: Intelligent traffic optimization using deep learning; awarded Rs. 2,500 by MMPSRPC, KSV.



- **4th Runner-Up – InnovAIItion Hackathon 2025**

Organized by Dhirubhai Ambani University.

Designed a multi-step real-time fraud detection pipeline using XGBoost and LSTM for accurate transaction analysis



- **Finalist – ANRF’s AISE Hack-2026** Organized by IBM and IIIT Hyderabad; fine-tuned the Prithvi-100M pretrained model using advanced transfer learning.